

ARA-Net: Atlas-Guided Multimodal Alzheimer’s Disease Staging with Locked External Subject-Level Validation and Structural Neurodegeneration Consistency

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Abstract

Background. The original ARA-Net submission was limited by weak external
classification evidence, an unsupported attention-based Clinical Alignment Score

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(CAS), and a non-significant Braak-stage analysis. We rebuilt the work as a subject-level, atlas-guided multimodal Alzheimer’s disease (AD) staging study with explicit external classification, error analysis, and bounded biological validation.

Methods. We constructed leakage-aware subject-level splits across ADNI, AIBL, IXI, and OASIS. ADNI was used for training, validation, and internal testing; AIBL was divided into adaptation training, adaptation validation, and a locked heldout external test; IXI served as a healthy negative-control cohort; and OASIS was retained only as an external stress test. The final model is a locked subject-level rescue probability ensemble that combines six base-model probability streams using log-probability pooling, class-specific offsets, temperature scaling, and subject-level probability averaging. The previous attention-only CAS claim was removed and replaced by atlas structural neurodegeneration consistency analysis in a priori AD-relevant regions.

Results. The original v3 model did not support the earlier external-generalization claim, with AIBL balanced accuracy (BAcc) of 0.399 and IXI healthy CN retention of 0.439. The final subject-level rescue ensemble achieved AIBL heldout accuracy of 0.903, BAcc of 0.833, macro AUC of 0.937, AD-vs-CN AUC of 1.000, and CN/MCI/AD recall of 0.961/0.686/0.852. Bootstrap 95% confidence intervals were 0.759–0.899 for BAcc, 0.531–0.839 for MCI recall, and 0.710–0.966 for AD recall. IXI healthy CN retention was 1.000. AIBL errors were concentrated near MCI/AD boundaries, with no AD subject misclassified as CN. OASIS transfer remained weak and is reported as a limitation. The AIBL heldout AD-key atlas-volume consistency score was 0.510 versus a uniform regional null of 0.286, with bootstrap CI 0.479–0.526 and permutation $p = 0.026$.

Conclusion. The revised work supports a domain-adapted, subject-level, atlas-guided multimodal AD staging framework with strong locked AIBL heldout performance, preserved IXI healthy specificity, and atlas-region structural neurodegeneration consistency. It does not claim pure zero-shot transfer, solved OASIS generalization, direct Braak-stage validation, or deployment-ready clinical performance.

Keywords: Alzheimer’s disease; structural MRI; atlas-guided multimodal learning; external validation; subject-level staging; neurodegeneration consistency; open-source research prototype

Abbreviations: AD, Alzheimer’s disease; ADNI, Alzheimer’s Disease Neuroimaging Initiative; AIBL, Australian Imaging, Biomarkers and Lifestyle study; AUC, area under the receiver-operating-characteristic curve; BAcc, balanced accuracy; CAS, Clinical Alignment Score; CN, cognitively normal; IXI, Information eXtraction from Images dataset; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination; OASIS, Open Access Series of Imaging Studies; sMRI, structural magnetic resonance imaging.

1 Introduction

Structural MRI is widely used in AD research because it captures neurodegeneration patterns including medial temporal atrophy, ventricular enlargement, and broader brain-volume change (Jack et al., 2018; Frisoni et al., 2010). Machine learning can use these signals to support CN/MCI/AD staging, but a model intended for scientific or translational use must demonstrate more than within-cohort classification performance. It must also show leakage-aware external evaluation, clear error behavior across clinically important classes, and appropriately bounded biological interpretation.

The original ARA-Net manuscript attempted to make explanation similarity and an attention-derived CAS central evidence for cross-dataset generalization and biological validity. The revised work deliberately changes that target. External generalization is evaluated directly through locked external classification, healthy negative-control testing, and subject-level endpoint aggregation. The old attention-only CAS is removed as a central biological claim. Because direct Braak-stage validation is unavailable and the original Braak correlation was non-significant, the biological claim is narrowed to a structural MRI neurodegeneration proxy.

We therefore rebuilt the study rather than making a narrow revision. The revised framework uses subject-level multi-cohort manifests, AIBL adaptation with a locked external heldout split, IXI healthy negative-control testing, OASIS stress testing, comparator models, bootstrap uncertainty, and MCI/AD error analysis. The final classifier is an atlas-guided multimodal probability ensemble that combines regional MRI features and core clinical variables, then averages repeated scans at subject level for the primary endpoint.

The revised contributions are:

- C1.** A leakage-aware subject-level protocol across ADNI, AIBL, IXI, and OASIS.
- C2.** Direct external CN/MCI/AD classification on a locked AIBL heldout endpoint.
- C3.** IXI healthy negative-control evaluation to quantify false impairment predictions.
- C4.** A locked probability rescue ensemble with class offsets, temperature scaling, and subject-level probability averaging.
- C5.** Bootstrap stability and subject-level MCI/AD error analysis.
- C6.** Replacement of the unsupported attention-only CAS/Braak claim with atlas structural neurodegeneration consistency analysis.
- C7.** An open-source research deployment package with explicit clinical-use boundaries.

Placeholder for Figure 1. Revised study framework and central model objective.

Panels to draw: **(a)** old v3 claim failure and V6 rebuild target; **(b)** subject-level cohort construction; **(c)** atlas-guided multimodal features; **(d)** locked rescue probability ensemble; **(e)** endpoint and claim-boundary summary. The visual emphasis should be that the paper is now an external subject-level AD staging framework, not an attention-only interpretability patch.

Figure 1: **Revised ARA-Net V6 framework.** The revised workflow uses leakage-aware subject-level cohort construction, atlas-guided multimodal feature extraction, a locked rescue probability ensemble, and subject-level probability averaging. The primary endpoint is locked AIBL heldout CN/MCI/AD staging; IXI is a healthy negative-control specificity endpoint; OASIS is retained as a stress-test limitation. The original attention-only CAS and direct Braak-stage claims are replaced by atlas structural neurodegeneration consistency analysis and explicit claim-boundary language.

Table 1: V6 split inventory. Counts refer to scan inventory and unique split-level subjects.

Split	Scans	Subjects	Role
ADNI train	1,686	450	Main training
ADNI validation	355	97	Validation and calibration
ADNI internal test	360	96	Internal risk analysis
AIBL adaptation training	719	385	External-domain adaptation
AIBL adaptation validation	191	105	External-domain calibration
AIBL locked heldout	397	210	Primary external test
OASIS external	99	99	Stress test
IXI healthy external	581	581	Healthy negative-control specificity

2 Methods

2.1 Cohorts and leakage-aware subject-level splits

All splits were defined at the subject level to reduce leakage from repeated scans (Varoquaux et al., 2017). ADNI was divided into training, validation, and internal test sets. AIBL was divided into adaptation training, adaptation validation, and a locked heldout external test. IXI was used as a healthy negative-control cohort. OASIS was retained as an external stress-test cohort and was not used for final model tuning.

Table 1 summarizes the split inventory. The subject counts in this table are unique participants used to define leakage-free cohort splits before endpoint aggregation. For longitudinal cohorts, the final endpoint units may differ from unique participants because repeated scans were averaged within subject-level diagnostic-state units.

2.2 Atlas-guided multimodal feature representation

MRI features were extracted from a 21-region atlas and included regional volumetric and intensity summaries. The main multimodal feature set combined atlas-derived MRI features with core clinical variables, including age, sex, education, APOE4, MMSE, and CDR-SB where available. Extended cognitive, biomarker, and volumetric clinical variables were used in sensitivity or comparator models rather than as the central scientific claim.

The revised model should be described as atlas-guided and multimodal. The primary scientific evidence is classification performance, healthy negative-control specificity, error analysis, and atlas structural neurodegeneration consistency.

2.3 Candidate models and locked rescue ensemble

The revised experimental framework included atlas-only, cascade, atlas+clinical, clinical-only, biomarker-enhanced, and ensemble models. The final locked model was a subject-level balanced rescue probability ensemble. It combined six base-model probability

Placeholder for Figure 2. Data line and endpoint design.

Panels to draw: **(a)** ADNI/AIBL/OASIS/IXI cohort blocks with scan and subject counts; **(b)** ADNI train-validation-internal-test split; **(c)** AIBL adaptation train/validation and locked heldout split; **(d)** endpoint aggregation from scans to subject-level diagnostic-state units; **(e)** OASIS explicitly shown as stress test, not tuning or success validation.

Figure 2: **Leakage-aware multi-cohort data line.** The revised study separates model development, external-domain adaptation, locked external testing, healthy-control specificity testing, and stress testing. This figure should make clear that AIBL heldout is the primary external endpoint, IXI evaluates false impairment in healthy controls, and OASIS is preserved only as a limitation stress test.

118 streams:

- 119 • AIBL-adapted atlas-biomarker-enhanced HGB.
- 120 • AIBL-adapted atlas-core-clinical HGB.
- 121 • AIBL-adapted clinical-biomarker-only RF balanced.
- 122 • AIBL-adapted clinical-core-only HGB.
- 123 • AIBL-adapted clinical-core-only RF balanced.
- 124 • RF-logistic regression ensemble component.

125 Let $p_{m,k}(x_i)$ denote the class probability for scan i , class k , and base model m , where
126 $k \in \{\text{CN}, \text{MCI}, \text{AD}\}$. The final ensemble uses log-probability pooling:

$$z_{i,k} = \frac{1}{T} \sum_{m=1}^M w_m \log(\max(p_{m,k}(x_i), \epsilon)) + b_k, \quad k \in \{\text{CN}, \text{MCI}, \text{AD}\}, \quad (1)$$

127 where $w_m \geq 0$, $\sum_m w_m = 1$, T is a temperature parameter, b_k is a class-specific offset,
128 and ϵ prevents numerical underflow. Calibrated class probabilities are:

$$\tilde{p}_{i,k} = \frac{\exp(z_{i,k})}{\sum_c \exp(z_{i,c})}. \quad (2)$$

129 For subject-level evaluation, repeated scans within subject-level diagnostic-state unit s
130 were averaged:

$$\bar{p}_{s,k} = \frac{1}{n_s} \sum_{i \in s} \tilde{p}_{i,k}, \quad \hat{y}_s = \arg \max_k \bar{p}_{s,k}. \quad (3)$$

131 2.4 Evaluation metrics, uncertainty, and error analysis

132 We report accuracy, balanced accuracy, macro one-vs-rest AUC, AD-vs-CN AUC, per-
133 class recall, precision, prediction distributions, and confusion matrices. For IXI, because
134 all subjects are healthy controls, the primary metric is CN retention, equivalent to one
135 minus the false impairment rate. For the locked AIBL heldout subject-level endpoint,
136 uncertainty was estimated using 2,000 bootstrap resamples.

137 MCI and AD errors were analyzed after subject-level probability averaging. We
138 summarized true/predicted transition rates and compared error groups by age, MMSE,
139 CDR-SB, APOE4, atlas hippocampal volume, atlas lateral ventricular volume, AD-like
140 atlas z-score, maximum predicted probability, and decision margin. The goal of this
141 analysis was to determine whether errors reflected complete collapse of disease classes
142 into CN or uncertainty near adjacent disease-stage boundaries.

Placeholder for Figure 3. Locked rescue ensemble model.

Panels to draw: **(a)** six base probability streams; **(b)** log-probability pooling with weights; **(c)** class offsets and temperature scaling; **(d)** scan-level to subject-level averaging; **(e)** final CN/MCI/AD probability output and decision. The main visual message is that the core model target is calibrated subject-level AD staging.

Figure 3: **Locked subject-level rescue probability ensemble.** The final model combines six base-model probability streams through log-probability pooling, class-specific offsets, temperature scaling, and subject-level probability averaging. This architecture is the central V6 model and should be shown as a calibrated staging system rather than a binary classifier or an attention-only network.

2.5 Structural neurodegeneration consistency

The old attention-centered CAS was removed because it did not provide valid biological evidence. The revised biological analysis instead tests whether disease-associated atlas volume changes concentrate in a priori AD-relevant regions: bilateral hippocampus, bilateral amygdala, and bilateral lateral ventricles. The score was compared against a uniform regional null using bootstrap confidence intervals and permutation testing.

This analysis is an MRI neurodegeneration proxy. It does not provide postmortem-stage validation or explanation-map biomarker evidence.

2.6 Open-source research deployment and clinical-use boundary

The public implementation includes reproducible analysis scripts, a research inference wrapper, an HTTP API/static frontend, aggregate reports, figures, documentation, and toy probability examples. Raw ADNI, AIBL, OASIS, and IXI data are governed by their original access agreements and are not redistributed.

The software is released as an open-source research prototype for retrospective evaluation and future prospective validation. It is not a medical device, is not cleared or approved for clinical use, and is not intended for standalone diagnosis or patient-care decisions.

3 Results

3.1 Original external failure and final external result

The original v3 model did not support the earlier cross-dataset generalization claim. On AIBL, it achieved accuracy 0.606, BAcc 0.399, and macro AUC 0.597. On IXI, only 0.439 of healthy controls were retained as CN, indicating a high false impairment rate.

The locked final rescue ensemble achieved substantially stronger AIBL heldout performance. At subject level, it achieved accuracy 0.903, BAcc 0.833, macro AUC 0.937, and AD-vs-CN AUC 1.000. CN/MCI/AD recall was 0.961/0.686/0.852. Bootstrap 95% confidence intervals were 0.759–0.899 for BAcc, 0.894–0.974 for macro AUC, 0.531–0.839 for MCI recall, and 0.710–0.966 for AD recall. The scan-level reference result was similar: AIBL heldout accuracy 0.909, BAcc 0.820, macro AUC 0.939, AD-vs-CN AUC 0.998, and CN/MCI/AD recall 0.964/0.642/0.854. On IXI, the final model retained 1.000 of healthy controls as CN at both scan and subject levels.

3.2 Comparator interpretation

The clinical-only RF comparator achieved AIBL heldout BAcc 0.835 and CN/MCI/AD recall 0.970/0.755/0.780 at scan level. This model is reported as a comparator and upper

Placeholder for Figure 4. External classification performance.

Panels to draw: **(a)** AIBL BAcc comparison: old v3, v4 atlas+clinical, final scan, final subject, clinical-only comparator; **(b)** AIBL macro AUC comparison; **(c)** final subject-level CN/MCI/AD recall; **(d)** IXI CN retention and false impairment rate; **(e)** OASIS as a small stress-test limitation panel. The central story is old external failure to new locked AIBL subject-level success with IXI specificity.

Figure 4: **External classification performance.** The original v3 model failed external classification and healthy specificity. The final subject-level rescue ensemble substantially improves locked AIBL heldout CN/MCI/AD staging and preserves IXI healthy specificity. OASIS is shown as a stress-test limitation, not as a successful validation cohort.

Table 2: Main external classification results. The locked primary endpoint is the final rescue ensemble at subject level on AIBL heldout.

Model/protocol	Unit	Test cohort	Endpoint n	Acc	BAcc	Macro AUC	AD-vs-CN AUC/CN retention	CN/MCI/AD recall	Role
Old v3 ensemble	scan	AIBL external	1,307	0.606	0.399	0.597	NA	NA	Failed external baseline
Old v3 ensemble	scan	IXI healthy	581	0.439	0.439	NA	CN retention 0.439	0.439/0.000/0.000	Failed specificity baseline
v4 atlas+clinical HGB	scan	AIBL heldout	397	0.882	0.741	0.942	AD-vs-CN AUC 0.990	0.964/0.528/0.732	Earlier rebuild
Final rescue ensemble	scan	AIBL heldout	397	0.909	0.820	0.939	AD-vs-CN AUC 0.998	0.964/0.642/0.854	Scan-level reference
Final rescue ensemble	subject	AIBL heldout	216	0.903	0.833	0.937	AD-vs-CN AUC 1.000	0.961/0.686/0.852	Locked primary result
Final rescue ensemble	subject	IXI healthy	581	1.000	1.000	NA	CN retention 1.000	1.000/0.000/0.000	Specificity check
Final rescue ensemble	subject	OASIS stress	99	0.586	0.334	0.554	AD-vs-CN AUC 0.371	0.966/0.034/0.000	Limitation
Clinical-only RF comparator	scan	AIBL heldout	397	0.922	0.835	0.957	AD-vs-CN AUC 0.997	0.970/0.755/0.780	Upper-bound comparator

Table 3: AIBL heldout subject-level confusion matrix for the locked final rescue ensemble.

True label	Predicted CN	Predicted MCI	Predicted AD	Recall
CN	148	5	1	0.961
MCI	2	24	9	0.686
AD	0	4	23	0.852

bound rather than the central ARA-Net model, because it does not retain the atlas-guided MRI component central to the revised scientific objective. This comparison is important: it shows that clinical variables carry substantial diagnostic signal and prevents overstating MRI-only interpretability.

3.3 Subject-level MCI and AD error analysis

On the locked AIBL heldout subject-level set, errors were concentrated at disease-stage boundaries. Among 154 CN endpoint units, 148 were classified as CN, five as MCI, and one as AD. Among 35 MCI endpoint units, 24 were classified as MCI, two as CN, and nine as AD. Among 27 AD endpoint units, 23 were classified as AD and four as MCI; no AD endpoint unit was misclassified as CN.

The feature-profile analysis supported a boundary-error interpretation. AIBL AD endpoint units correctly classified as AD had lower MMSE, larger lateral ventricular volume, and higher AD-like atlas z-scores than AD endpoint units classified as CN/MCI. MCI endpoint units classified as AD had lower MMSE and more AD-like atlas profiles than MCI endpoint units classified correctly, consistent with a disease-severity boundary rather than arbitrary failure.

3.4 Bootstrap stability

Bootstrap analysis with 2,000 resamples quantified uncertainty for the locked AIBL heldout subject-level endpoint. The 95% CI for BAcc was 0.759–0.899. The 95% CI for MCI recall was 0.531–0.839, and the 95% CI for AD recall was 0.710–0.966. These intervals support the final model as the locked primary result while preserving MCI staging uncertainty as a visible limitation.

Placeholder for Figure 5. Subject-level confusion and error profile.

Panels to draw: **(a)** AIBL heldout subject-level confusion heatmap; **(b)** internal subject-level confusion heatmap; **(c)** AIBL transition/alluvial flow; **(d)** MCI-to-CN, MCI-to-AD, AD-to-MCI, AD-to-CN error subtype bars; **(e)** error-feature profile summary using MMSE, ventricle volume, hippocampus volume, AD-like z-score, confidence, and margin. The key visual claim is zero AD-to-CN errors on AIBL and boundary-like residual mistakes.

Figure 5: **Subject-level MCI/AD error behavior.** AIBL heldout errors concentrate near adjacent disease-stage boundaries rather than collapsing disease cases into CN. Internal calibration remains weaker and should be interpreted as a limitation.

Placeholder for Figure 6. Stability and uncertainty.

Panels to draw: **(a)** bootstrap BAcc distribution with 95% CI; **(b)** bootstrap MCI recall distribution; **(c)** bootstrap AD recall distribution; **(d)** forest plot of BAcc, macro AUC, MCI recall, AD recall; **(e)** small note that OASIS was excluded from tuning. The visual message is that the final endpoint is stable enough to lock, with MCI as the main uncertainty.

Figure 6: **Bootstrap stability of the locked primary endpoint.** The final AIBL subject-level result was evaluated with 2,000 bootstrap resamples. The distributions and confidence intervals support the locked model while identifying MCI recall as the main residual uncertainty.

3.5 Structural neurodegeneration consistency and OASIS stress test

The AIBL heldout atlas-volume consistency score exceeded the uniform regional null in the a priori AD-key regions. The score was 0.510 versus a uniform null of 0.286, with error-difference 0.225, bootstrap CI 0.479–0.526, and permutation $p = 0.026$.

Across all labeled AD-relevant data, the AD-key consistency score was 0.426 versus a uniform null of 0.286, with permutation $p = 0.0207$. The ADNI-only internal check remained non-significant, with score 0.342 and $p = 0.1843$. These results support a bounded structural MRI neurodegeneration proxy while making clear that postmortem-stage validation is not available.

OASIS remained weak without OASIS tuning. The final subject-level model achieved OASIS accuracy 0.586, BAcc 0.334, macro AUC 0.554, AD-vs-CN AUC 0.371, and CN/MCI/AD recall 0.966/0.034/0.000. This result is reported as an unresolved transfer limitation, not as successful external validation.

4 Discussion

The revised study addresses the major weaknesses of the original manuscript by changing both the evidence base and the claim boundary. Cross-dataset generalization is no longer inferred from explanation similarity. It is evaluated using a locked AIBL heldout subject-level test and an IXI healthy negative-control cohort. The unsupported attention-centered CAS claim is removed and replaced by an atlas-region structural neurodegeneration consistency analysis. The non-significant disease-stage result is no longer used to claim direct pathological staging.

The final model is strongest when interpreted as a domain-adapted external AD staging framework. AIBL adaptation data were used for model fitting and calibration, but AIBL heldout endpoint units remained locked and were not used for final endpoint evaluation. This distinction is important: the work does not solve pure ADNI-to-AIBL zero-shot staging, but it does demonstrate that anatomically grounded MRI features and core clinical variables can support robust heldout subject-level staging within an external cohort.

The error analysis clarifies the clinical meaning of the remaining failures. On AIBL heldout, AD endpoint units were not missed as CN; residual AD errors were classified as MCI. MCI errors were split between correct MCI and AD, with only two MCI endpoint units classified as CN. This pattern is preferable to a model that preserves overall accuracy by collapsing minority disease classes into CN, but it also shows that precise MCI/AD boundary staging remains challenging.

The biological analysis is intentionally bounded. The AIBL heldout atlas-volume

Placeholder for Figure 7. CAS/Braak replacement and biological consistency.

Panels to draw: **(a)** old attention-only CAS failure callout; **(b)** definition of AD-key atlas regions: bilateral hippocampus, amygdala, and lateral ventricles; **(c)** AIBL heldout structural consistency score 0.510 vs null 0.286 with CI and $p = 0.026$; **(d)** pooled all-labeled-AD score 0.426 vs null 0.286, plus ADNI-only non-significant internal check; **(e)** claim-boundary badges: MRI proxy only, not attention biomarker, not direct Braak proof, not clinical device.

Figure 7: Biological consistency and replacement of unsupported CAS/Braak claims. The invalid attention-only CAS claim is removed. The revised biological analysis tests whether atlas structural MRI changes concentrate in a priori AD-relevant regions. The supported claim is a bounded MRI neurodegeneration proxy, not attention-map biomarker discovery or direct Braak-stage validation.

consistency result supports disease-consistent structural MRI change in AD-relevant regions. It does not demonstrate explanation-map biomarkers, and it does not provide postmortem-stage validation. This narrower claim is more defensible and better aligned with the available data.

Several limitations remain. OASIS transfer is not solved and should be treated as a stress-test failure requiring larger and cleaner external cohorts. The internal subject-level calibration pattern remains modest, especially for CN specificity within the internal split. Clinical variables contain substantial diagnostic signal, as shown by the clinical-only comparator. The 21-region atlas is anatomically interpretable but coarse, and finer parcellations may better capture cortical AD patterns. Finally, although AIBL heldout performance is strong, the model is not presented as deployment-ready clinical software; it is best framed as an open-source research prototype requiring prospective validation.

5 Conclusion

This revised work is a substantive rebuild of the original ARA-Net study. It replaces unsupported attention-centered claims with locked external subject-level classification, healthy negative-control specificity, bootstrap uncertainty, MCI/AD error analysis, and a bounded atlas structural neurodegeneration consistency analysis. The strongest supported claim is domain-adapted external AD staging with an MRI neurodegeneration proxy, not pure zero-shot generalization, postmortem-stage validation, OASIS success, or clinical deployment readiness.

Data Availability

Raw ADNI, AIBL, OASIS, and IXI data are governed by their original data-use agreements and are not redistributed. Public repository files are limited to code, aggregate reports, figures, documentation, final model configuration, and toy probability examples. Row-level subject/scan predictions, private clinical spreadsheets, MRI volumes, and model checkpoints are excluded from the public package.

Code Availability

The public research implementation, documentation, aggregate reports, deployment wrappers, and final ensemble configuration are available at <https://github.com/Lava168/ARA-Net>. The released software is a research prototype; it is not a medical device and is not cleared or approved for clinical use.

Author Contributions (CRediT)

Yuanqin Zhao: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing – Original Draft, Visualization. **Yu Zhang:** Methodology, Software, Validation, Writing – Review & Editing. **Jinghong Hu:** Investigation, Data Curation, Writing – Review & Editing. **Jian Mao:** Methodology, Validation, Writing – Review & Editing. **Bisong Zhou:** Data Curation, Writing – Review & Editing. **Zhuoyao Song:** Data Curation, Writing – Review & Editing. **Tengfei Guo:** Conceptualization, Supervision, Funding Acquisition, Writing – Review & Editing. **Shaohua Ma:** Conceptualization, Supervision, Funding Acquisition, Project Administration, Writing – Review & Editing.

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Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- World Health Organization, Dementia fact sheet, 2023. <https://www.who.int/news-room/fact-sheets/detail/dementia>
- C.R. Jack, D.A. Bennett, K. Blennow, M.C. Carrillo, B. Dunn, S.B. Haeberlein, et al., NIA-AA research framework: toward a biological definition of Alzheimer’s disease, *Alzheimer’s & Dementia* 14(4) (2018) 535–562. <https://doi.org/10.1016/j.jalz.2018.02.018>
- S. Korolev, A. Safiullin, M. Belyaev, Y. Dodonova, Residual and plain convolutional neural networks for 3D input volumes, arXiv preprint arXiv:1701.06643, 2017.
- A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, et al., An image is worth 16x16 words: transformers for image recognition at scale, in: ICLR, 2020.
- H. Jia, et al., STNet: spatial-temporal network for Alzheimer’s disease diagnosis, in: Proceedings of MICCAI, Springer, 2023.
- J. Gao, et al., Robust LSTM-based classification for Alzheimer’s disease, in: Proceedings of IEEE EMBC, 2023.
- M. Alkhatami, et al., ECAResNet with SMOTE and focal loss for Alzheimer’s disease classification, *Biomedical Signal Processing and Control* 99 (2025) 106882. <https://doi.org/10.1016/j.bspc.2024.106882>
- S. Murugan, et al., DEMNET: a deep learning model for early diagnosis of Alzheimer’s diseases and dementia from MRI images, *Biomedical Signal Processing and Control* 65 (2021) 102386. <https://doi.org/10.1016/j.bspc.2020.102386>
- A. Majee, et al., 3D HCCT: a hybrid CNN-convolutional transformer for brain MRI classification, *Pattern Recognition* 148 (2024) 110197. <https://doi.org/10.1016/j.patcog.2023.110197>
- X. Zhang, et al., THAN: task-driven hierarchical attention network for the diagnosis of Alzheimer’s disease, *NeuroImage* 253 (2022) 119099. <https://doi.org/10.1016/j.neuroimage.2022.119099>
- R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-CAM: visual explanations from deep networks via gradient-based localization, in: Proceedings of ICCV, IEEE, 2017, pp. 618–626.
- S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems (NeurIPS), 2017.

326 M. Sundararajan, A. Taly, Q. Yan, Axiomatic attribution for deep networks, in: Proceed-
327 ings of ICML, 2017.

328 J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, B. Kim, Sanity checks for
329 saliency maps, in: Advances in Neural Information Processing Systems (NeurIPS),
330 2018.

331 C. Rudin, Stop explaining black box machine learning models for high stakes decisions
332 and use interpretable models instead, *Nature Machine Intelligence* 1 (5) (2019) 206–215.
333 <https://doi.org/10.1038/s42256-019-0048-x>

334 J. Amann, A. Blasimme, E. Vayena, D. Frey, V.I. Madai, Explainability for artificial
335 intelligence in healthcare: a multidisciplinary perspective, *BMC Medical Informatics*
336 and Decision Making 20 (2020) 310. <https://doi.org/10.1186/s12911-020-01332-6>

337 G. Varoquaux, P.R. Raamana, D.A. Engemann, A. Hoyos-Idrobo, Y. Schwartz, B.
338 Thirion, Assessing and tuning brain decoders: cross-validation, caveats, and guide-
339 lines, *NeuroImage* 145 (2017) 166–179. [https://doi.org/10.1016/j.neuroimage.](https://doi.org/10.1016/j.neuroimage.2016.10.038)
340 [2016.10.038](https://doi.org/10.1016/j.neuroimage.2016.10.038)

341 H. Braak, E. Braak, Neuropathological staging of Alzheimer-related changes, *Acta Neu-*
342 *ropathologica* 82 (4) (1991) 239–259. <https://doi.org/10.1007/BF00308809>

343 L. Henschel, S. Conjeti, S. Estrada, K. Diers, B. Fischl, M. Reuter, FastSurfer—a fast and
344 accurate deep learning based neuroimaging pipeline, *NeuroImage* 219 (2020) 117012.
345 <https://doi.org/10.1016/j.neuroimage.2020.117012>

346 Z. Zhou, V. Sodha, J. Pang, M.B. Gotway, J. Liang, Models Genesis: generic autodidactic
347 models for 3D medical image analysis, *Medical Image Analysis* 67 (2021) 101840. <https://doi.org/10.1016/j.media.2020.101840>

348 <https://doi.org/10.1016/j.media.2020.101840>

349 I. Loshchilov, F. Hutter, Decoupled weight decay regularization, in: Proceedings of ICLR,
350 2019.

351 H. Zhang, M. Cisse, Y.N. Dauphin, D. Lopez-Paz, mixup: beyond empirical risk mini-
352 mization, in: Proceedings of ICLR, 2018.

353 European Commission, Proposal for a regulation laying down harmonised rules on artifi-
354 cial intelligence (AI Act), COM/2021/206, 2021.

355 S.P. Poulin, R. Dautoff, J.C. Morris, L.F. Barrett, B.C. Dickerson, Amygdala atro-
356 phy is prominent in early Alzheimer’s disease and relates to symptom severity, *Psy-*
357 *chiatry Research: Neuroimaging* 194 (1) (2011) 7–13. [https://doi.org/10.1016/j.](https://doi.org/10.1016/j.psychresns.2011.06.014)
358 [psychresns.2011.06.014](https://doi.org/10.1016/j.psychresns.2011.06.014)

359 G. Livingston, J. Huntley, A. Sommerlad, D. Ames, C. Ballard, S. Banerjee, et al., De-
 360 mentia prevention, intervention, and care: 2020 report of the Lancet Commission,
 361 The Lancet 396 (10248) (2020) 413–446. [https://doi.org/10.1016/S0140-6736\(20\)](https://doi.org/10.1016/S0140-6736(20)30367-6)
 362 [30367-6](https://doi.org/10.1016/S0140-6736(20)30367-6)

363 B. Dubois, H. Hampel, H.H. Feldman, P. Scheltens, P. Aisen, S. Andrieu, et al., Preclinical
 364 Alzheimer’s disease: definition, natural history, and diagnostic criteria, Alzheimer’s &
 365 Dementia 12 (3) (2016) 292–323. <https://doi.org/10.1016/j.jalz.2016.02.002>

366 R.C. Petersen, O. Lopez, M.J. Armstrong, T.S.D. Getchius, M. Ganguli, D. Gloss,
 367 et al., Practice guideline update summary: mild cognitive impairment, Neurology 90 (3)
 368 (2018) 126–135. <https://doi.org/10.1212/WNL.0000000000004826>

369 G.B. Frisoni, N.C. Fox, C.R. Jack, P. Scheltens, P.M. Thompson, The clinical use of
 370 structural MRI in Alzheimer disease, Nature Reviews Neurology 6 (2) (2010) 67–77.
 371 <https://doi.org/10.1038/nrneurol.2009.215>

372 J. Wen, E. Thibeau-Sutre, M. Diaz-Melo, J. Samper-González, A. Routier, S. Bottani,
 373 et al., Convolutional neural networks for classification of Alzheimer’s disease: overview
 374 and reproducible evaluation, Medical Image Analysis 63 (2020) 101694. [https://doi.](https://doi.org/10.1016/j.media.2020.101694)
 375 [org/10.1016/j.media.2020.101694](https://doi.org/10.1016/j.media.2020.101694)

376 T. Jo, K. Nho, A.J. Saykin, Deep learning in Alzheimer’s disease: diagnostic classification
 377 and prognostic prediction using neuroimaging data, Frontiers in Aging Neuroscience 11
 378 (2019) 220. <https://doi.org/10.3389/fnagi.2019.00220>

379 G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional
 380 networks, in: Proceedings of CVPR, IEEE, 2017, pp. 2261–2269.

381 Z.C. Lipton, The mythos of model interpretability, Queue 16 (3) (2018) 31–57. [https:](https://doi.org/10.1145/3236386.3241340)
 382 [//doi.org/10.1145/3236386.3241340](https://doi.org/10.1145/3236386.3241340)

383 R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, D. Pedreschi, A survey
 384 of methods for explaining black box models, ACM Computing Surveys 51 (5) (2019)
 385 93:1–93:42. <https://doi.org/10.1145/3236009>

386 A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, et al., Attention
 387 is all you need, in: Advances in Neural Information Processing Systems (NeurIPS),
 388 2017, pp. 5998–6008.

389 D. Bahdanau, K. Cho, Y. Bengio, Neural machine translation by jointly learning to align
 390 and translate, in: Proceedings of ICLR, 2015.

391 B. Fischl, FreeSurfer, NeuroImage 62 (2) (2012) 774–781. [https://doi.org/10.1016/j.](https://doi.org/10.1016/j.neuroimage.2012.01.021)
 392 [neuroimage.2012.01.021](https://doi.org/10.1016/j.neuroimage.2012.01.021)

- N. Tzourio-Mazoyer, B. Landeau, D. Papathanassiou, F. Crivello, O. Etard, N. Delcroix, et al., Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain, *NeuroImage* 15 (1) (2002) 273–289. <https://doi.org/10.1006/nimg.2001.0978>
- R.S. Desikan, F. Ségonne, B. Fischl, B.T. Quinn, B.C. Dickerson, D. Blacker, et al., An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest, *NeuroImage* 31 (3) (2006) 968–980. <https://doi.org/10.1016/j.neuroimage.2006.01.021>
- S. Parisot, S.I. Ktena, E. Ferrante, M. Lee, R. Guerrero, B. Glocker, D. Rueckert, Disease prediction using graph convolutional networks: application to autism spectrum disorder and Alzheimer’s disease, *Medical Image Analysis* 48 (2018) 117–130. <https://doi.org/10.1016/j.media.2018.06.001>
- X. Li, Y. Zhou, N. Dvornek, M. Zhang, S. Gao, J. Zhuang, et al., BrainGNN: interpretable brain graph neural network for fMRI analysis, *Medical Image Analysis* 74 (2021) 102233. <https://doi.org/10.1016/j.media.2021.102233>
- C.R. Jack, M.A. Bernstein, N.C. Fox, P. Thompson, G. Alexander, D. Harvey, et al., The Alzheimer’s Disease Neuroimaging Initiative (ADNI): MRI methods, *Journal of Magnetic Resonance Imaging* 27 (4) (2008) 685–691. <https://doi.org/10.1002/jmri.21049>
- D.S. Marcus, T.H. Wang, J. Parker, J.G. Csernansky, J.C. Morris, R.L. Buckner, Open Access Series of Imaging Studies (OASIS): cross-sectional MRI data in young, middle aged, nondemented, and demented older adults, *Journal of Cognitive Neuroscience* 19 (9) (2007) 1498–1507. <https://doi.org/10.1162/jocn.2007.19.9.1498>
- J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., Lawrence Erlbaum Associates, Hillsdale, NJ, 1988.
- P. Scheltens, K. Blennow, M.M.B. Breteler, B. de Strooper, G.B. Frisoni, S. Salloway, W.M. Van der Flier, Alzheimer’s disease, *The Lancet* 388(10043) (2016) 505–517. [https://doi.org/10.1016/S0140-6736\(15\)01124-1](https://doi.org/10.1016/S0140-6736(15)01124-1)
- B.C. Dickerson, I. Goncharova, M.P. Sullivan, C. Forchetti, R.S. Wilson, D.A. Bennett, et al., MRI-derived entorhinal and hippocampal atrophy in incipient and very mild Alzheimer’s disease, *Neurobiology of Aging* 22 (5) (2001) 747–754. [https://doi.org/10.1016/S0197-4580\(01\)00271-8](https://doi.org/10.1016/S0197-4580(01)00271-8)
- A. Delacourte, J.P. David, N. Sergeant, L. Buée, A. Wattez, P. Vermersch, et al., The biochemical pathway of neurofibrillary degeneration in aging and Alzheimer’s disease, *Neurology* 52 (6) (1999) 1158–1165. <https://doi.org/10.1212/WNL.52.6.1158>

- J.-P. Fortin, D. Parker, B. Tunc, T. Watanabe, M.A. Elliott, K. Ruparel, et al., Harmonization of multi-site diffusion tensor imaging data, *NeuroImage* 161 (2017) 149–170. <https://doi.org/10.1016/j.neuroimage.2017.08.047>
- R. Pomponio, G. Erus, M. Habes, H. Doshi, D. Srinivasan, E. Mamourian, et al., Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan, *NeuroImage* 208 (2020) 116450. <https://doi.org/10.1016/j.neuroimage.2019.116450>
- E. Moradi, A. Pepe, C. Gaser, H. Huttunen, J. Tohka, Machine learning framework for early MRI-based Alzheimer’s conversion prediction in MCI subjects, *NeuroImage* 104 (2015) 398–412. <https://doi.org/10.1016/j.neuroimage.2014.10.002>
- C.R. Jack, D.A. Bennett, K. Blennow, M.C. Carrillo, H.H. Feldman, G.B. Frisoni, et al., A/T/N: an unbiased descriptive classification scheme for Alzheimer disease biomarkers, *Neurology* 91 (3) (2018) 111–116. <https://doi.org/10.1212/WNL.0000000000005560>
- E.H. Corder, A.M. Saunders, W.J. Strittmatter, D.E. Schmechel, P.C. Gaskell, G.W. Small, et al., Gene dose of apolipoprotein E type 4 allele and the risk of Alzheimer’s disease in late onset families, *Science* 261 (5123) (1993) 921–923. <https://doi.org/10.1126/science.8346443>
- A. Schaefer, R. Kong, E.M. Gordon, T.O. Laumann, X.-N. Zuo, A.J. Holmes, et al., Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI, *Cerebral Cortex* 28 (9) (2018) 3095–3114. <https://doi.org/10.1093/cercor/bhx179>